

YASH PARSAI

Machine Learning Engineer | Data Scientist

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 8 months of hands-on experience at TeamLease RegTech building predictive systems, classification models, risk-scoring frameworks, and scalable data pipelines. Proficient in Python, Scikit-learn, TensorFlow, XGBoost, SQL, and ClickHouse across the ML lifecycle — data ingestion, ETL, feature engineering, model training, hyperparameter optimization (GridSearchCV), and evaluation. Actively working with open-source LLMs, Agentic AI workflows (LangChain, tool calling, multi-step agents), GPU inference, and efficient model serving. Published IGI Global researcher (2026) in deep learning / OCR. Seeking Machine Learning Engineer or Data Scientist roles.

CORE COMPETENCIES

Machine Learning • Data Science • Statistical Modeling • Feature Engineering • Model Evaluation • ETL / Data Pipelines • Deep Learning • NLP • LLMs • Agentic AI • AI Agents • Inference Optimization • MLOps

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern | TeamLease RegTech

May 2025 – Dec 2025 • Remote

Python • Scikit-learn • XGBoost • GridSearchCV • Power BI • MySQL • MSSQL • ClickHouse • ETL

- Built an AI-powered compliance smart-notification system using login behavior, compliance history, and submission patterns across **10,000+ user accounts**, reducing manual follow-up effort for compliance managers.
- Improved escalation prediction accuracy by **18%** through systematic feature engineering, GridSearchCV hyperparameter tuning, and k-fold cross-validation on supervised classification models.
- Designed a compliance risk-scoring framework mapping penalty severity and submission-delay patterns to user performance metrics; outputs used in executive risk reporting.
- Engineered scalable Python ETL pipelines integrating **MySQL, MSSQL, and ClickHouse** — covering ingestion, cleaning, transformation, and aggregation for 10,000+ records. Structured pipelines for scheduled execution and multi-source consistency; delivered Power BI dashboards for operational and executive stakeholders.
- Identified repeat non-compliant entities via late-submission-rate analysis and cross-company behavior tracking, generating actionable insights adopted by the compliance operations team.

PUBLICATION

[Book Chapter](#) — **IGI Global (2026)**: AI-Driven Handwritten Text Recognition and Digital Conversion for a Smart Campus. In *Reinventing Higher Education Through Digital Transformation and Smart Campuses*, IGI Global. DOI: 10.4018/979-8-3373-6831-3.ch002

KEY PROJECTS

IPO Readiness Assessment Tool • Python, XGBoost, KMeans, FinBERT, LangChain, Gemini API, Streamlit, Plotly, PostgreSQL

- End-to-end IPO scoring engine across 6 financial dimensions. Integrated XGBoost risk classification, KMeans peer clustering, and FinBERT sentiment analysis using LangChain + Gemini API (tool calling & structured outputs); deployed interactive Streamlit + Plotly dashboards.

Handwritten Text Recognition (CRNN + OCR) • Python, TensorFlow, Keras, OpenCV, CNN, BiLSTM, CTC Loss

- Trained CRNN (CNN + BiLSTM + CTC Loss) on **90,000+ labeled samples** → **87% word-level accuracy**; reduced CER by **22%** via preprocessing and augmentation. Published as IGI Global book chapter (2026).

AI Travel Itinerary Planner • Python, FastAPI, PostgreSQL + PostGIS, OR-Tools, Docker, Mapbox/Leaflet

- Full-stack platform with FastAPI backend, PostGIS-backed attractions schema (city-scoped IDs, clusters, must-visit ranking), and TSP route optimization (OR-Tools). Designed data model and ingestion flow for multi-city attractions data.

Water Quality Prediction System • Python, Scikit-learn, Random Forest, Gradient Boosting, Pandas, NumPy

- Multi-output regression (WQI, PIG, EWQI) from 11+ parameters; **$R^2 = 0.89$** with Random Forest (outperformed Gradient Boosting). Built automated evaluation pipeline for rapid model comparison.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, HTML, Bash

ML & Modeling: Scikit-learn, XGBoost, LightGBM, Random Forest, Logistic Regression, SVM, K-Means, GridSearchCV, Cross-Validation, Feature Engineering, Model Evaluation (Accuracy, Precision, Recall, F1, R^2 , CER)

Deep Learning & LLMs: TensorFlow, Keras, PyTorch, CNN, LSTM, BiLSTM, CRNN, Hugging Face Transformers, Open-source LLMs, LoRA / PEFT, Quantization, GPU Inference

Agentic AI: LangChain, LangGraph, Tool Calling, AI Agents, Multi-Agent Workflows, ReAct, Structured Outputs, Gemini API

Data Engineering: Pandas, NumPy, ETL Pipelines, Multi-source Integration (MySQL, MSSQL, ClickHouse), Apache Airflow, PySpark, dbt

Databases: MySQL, MSSQL, ClickHouse, PostgreSQL, PostGIS

MLOps & Tools: Docker, FastAPI, MLflow, Git, Streamlit, AWS S3, Azure ML Studio

Visualization: Power BI, Matplotlib, Seaborn, Plotly, Streamlit

EDUCATION

B.Tech – Computer Science Engineering | Jaypee University of Engineering & Technology (JUET), Guna, MP

CGPA: 7.5 / 10

ACHIEVEMENTS & CERTIFICATIONS

- **Shortlisted — Amazon ML Summer School 2026** (top 3,000 of applicant pool, selected via Amazon University Talent Acquisition, India)
- [AWS](#) — ML Foundations, ML Terminology & Process, Planning a Machine Learning Project (2026)
- [ISRO](#) — Geodata Processing using Python & ML; START Program; Method of Space Science Research
- [Microsoft](#) — AI in Action
- Additional General Secretary, JYC, JUET
- CSE Technical Clubs Head, JYC, JUET
- Co-Organizer, TEDxJUET — speaker coordination, logistics & cross-functional execution